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RESEARCH INTERESTS

Representation learning, Watermarking, Multimodality

EDUCATION

University of Nebraska Omaha, Omaha, Nebraska, USA

Ph.D., Computing & Information Science

2021 – 2026

Dissertation: *Invariant Feature Learning in AI Models: Exploring Representation Spaces for Robust Watermarking and Beyond* (Dr. Xin Zhong)

Utah State University, Logan, Utah, USA

M.S., Computer Science

2018 – 2020

Thesis: *Deep Q-Learning Applied to Stock Trading* (Dr. Nicholas Flann)

West Bengal University of Technology, Kolkata, India

B.Tech., Computer Science

2013 – 2017

RESEARCH EXPERIENCE

University of Nebraska Omaha, Graduate Researcher

2021 – 2026

Representation learning, watermarking, and LLM research.

Utah State University, Graduate Researcher

2018 – 2020

Deep reinforcement learning research for financial trading applications.

PUBLICATIONS

Conference Publications

- [C1] A. A. Tanvir, **A. Dasgupta**, X. Zhong. **TIACam: Text-Anchored Invariant Feature Learning with Auto-Augmentation for Camera-Robust Zero-Watermarking.** 2026
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026.
(in press)
- [C2] H. Yoon, X. Zhong, **A. Dasgupta**, G. Nugent, G. Trainin. **Leveraging Artificial Intelligence (AI) to Enhance Computer Science Instruction.** 2024
IEEE Frontiers in Education Conference (FIE), 2024.
- [C3] **A. Dasgupta**, X. Zhong. **Robust Image Watermarking Based on Cross-Attention and Invariant Domain Learning.** 2023
International Conference on Computational Science and Computational Intelligence (CSCI), 2023.
- [C4] N. Khatri, **A. Dasgupta**, Y. Shen, X. Zhong, F. Y. Shih. **Perspective Transformation Layer.** 2022
International Conference on Computational Science and Computational Intelligence (CSCI), 2022.

Journal Publications

- [J1] **A. Dasgupta**, A. A. Tanvir, X. Zhong. **Watermarking Language Models through Language Models.** 2025
IEEE Transactions on Artificial Intelligence, pp. 1–10, 2025.
DOI: 10.1109/TAI.2025.3605117

Book Chapters

- [B1] **A. Dasgupta, X. Zhong. Enhanced image watermarking through cross-attention and noise-invariant domain learning.** *Book Chapter, Imaging Science: Computer Vision, Image and Signal Processing, Pattern Recognition, IEEE*, 2024.

Submitted / Under Review

- [S1] **A. Dasgupta, A. A. Tanvir, X. Zhong. Invariant Features in Language Models: Geometric Characterization and Model Attribution.** Submitted to *NeurIPS 2026*.

THESES AND OTHER RESEARCH

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| [O1] | Retinal Vascular Leakage Detection
• Trained classifiers on patches of ultra-wide-field fluorescein angiography scans to detect areas of leakage, in collaboration with medical practitioners. Presented at the American Academy of Ophthalmology (AAO), 2025. | 2025 |
| [O2] | Sentiment Analysis on Electric Vehicle Discourse
• Analyzed public sentiment using lexical and Word2Vec-LSTM approaches on scraped Twitter data. | 2022 |
| [O3] | Deep Q-Learning Applied to Stock Trading
• Designed and evaluated a reinforcement learning trading system using auto-scraped S&P 500 index data as part of the M.S. thesis. | 2020 |

TEACHING EXPERIENCE

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| University of Nebraska Omaha , Graduate Teaching Assistant
Courses: Introduction to Python, Capstone
• Assisted in assignment design, held office hours, and provided guidance on project scope, feasibility, and implementation strategy for senior capstone projects. | 2023–2026 |
| University of Nebraska Omaha , Intern Supervisor
• Interviewed and shortlisted high school intern candidates for AI research positions at the RNA Lab.
• Supervised interns on active research projects in representation learning and computer vision, adapting technical content for students with varying levels of prior knowledge. | 2022–2023 |
| Utah State University , Graduate Teaching Assistant
Courses: Programming Languages, Introduction to C++, Introduction to AI
• Assisted in designing assignments with progressive difficulty structures across three courses.
• Held weekly office hours and provided feedback on implementation and conceptual understanding. | 2018–2020 |

GRANTS AND AWARDS

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| • Graduate Research and Creative Activity (GRACA) Fellowship , \$10,000
University of Nebraska Omaha | 2021, 2025 |
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REVIEWER SERVICE

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| • ACCV, Scientific Reports (Nature), Springer Nature | 2022–PRESENT |
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SOFTWARE AND CODE

LLMID (2026) — <https://github.com/cent664/LLMID>

A geometric framework for identifying invariant subspaces in pretrained LLMs across 9 open-source model families, without additional training. The framework applies contrastive sensitivity decomposition to separate semantic and nuisance directions in hidden representations, validated through causal patching experiments. The discovered invariant zones achieve 90%+ zero-shot model attribution accuracy across base, fine-tuned, and distilled model variants.

TIACam (2026) — WIP

A camera-robust zero-watermarking framework that combines CLIP image-text features with a learned adversarial augmentation module to ground invariant representations in semantic content. The auto-augmentor discovers camera-like distortions through differentiable operators, removing the need for manually designed augmentation families. Achieves 95%+ watermark recovery on ImageNet and Visual Genome datasets under real-world camera distortions.

LLMWM (2025) — <https://github.com/cent664/LLMWM>

A model-agnostic prompt-based watermarking framework for LLM-generated text that operates without access to model weights or decoding internals. The pipeline was evaluated across 25 open-source LLMs including fine-tuned and distilled variants trained using LoRA and DeepSpeed. Achieves 88%+ detection accuracy under paraphrasing, fine-tuning, and adversarial prompt attacks.

SSRIW (2023) — <https://github.com/cent664/SSRIW>

A ViT-based cross-attention watermarking system that directly trains an invariant domain alongside the watermarking task using a triplet loss objective. The cross-attention mechanism localizes semantically relevant regions for watermark embedding, and the invariant domain provides robustness under compound distortions. Achieves state-of-the-art comparable accuracy under common noise types with improved performance under combined augmentations.

PTL (2022) — https://github.com/kcnishan/Perspective_Transformation_Layer

A lightweight differentiable layer that learns 2D projections of 3D views for multiple viewpoints, implemented as a drop-in module within standard deep learning pipelines. Rather than relying on fixed augmentation pipelines, PTL internalizes perspective warping as a learnable operator, allowing the model to adaptively select viewpoint transformations that improve downstream task accuracy.

TECHNICAL SKILLS

ML CONCEPTS:	Representation Learning, Watermarking, Computer Vision, NLP, LLMs
FRAMEWORKS:	PyTorch, TensorFlow, HuggingFace Transformers, scikit-learn
PROGRAMMING:	Python, Bash, CUDA, Git, Linux
DISTRIBUTED:	DeepSpeed, LoRA, PyTorch DDP, Slurm/HPC